

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2023/2024

MAY/JUNE EXAMINATION

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

THURSDAY, 30 MAY 2024

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

Question 1

Figure 1-1: Google LLC's autonomous car

Since its launching in USA in 2007, Google's Street View cars (see **Figure 1-1**) have been circling the globe, using roof-mounted cameras to map more than 10 million miles of imagery. As the team lead of Google LLC's autonomous car project, you are working very hard to convert all its manual driving cars to automated driving cars, which are referred to as the agent-X. Suppose multiple versions of agent-X are implemented for experimental purposes based on five different agent design programs, which are *table-driven agent*, *simple reflex agent*, *model-based reflex agent*, *goal-based agent*, *utility-based agent*, answer the following questions:

- a) Describe in your own words what it means by *agent*. (2 marks)
- b) A *rational agent* is a type of intelligent agent that acts based on the concept of *rationality*. Briefly describe the **FOUR (4)** components of *rationality*. Then, describe in your own words what it means by *rational agent*. (4 + 2 marks)
- c) Explain how a *simple reflex agent* works with the aid of an *illustration*. (4 marks)
- d) Identify **THREE (3)** challenges of building a *utility-based agent*. (3 marks)
- e) Identify an agent program between *table-driven agent*, and *simple reflex agent* that is more suitable to be applied as the solution. Explain why your selected choice is more suitable and why another agent program is less suitable to support your answer. (1 + 4 marks)
- f) Identify an agent program between *goal-based agent*, and *utility-based agent* that is more suitable to be applied as the solution. Explain why your selected choice is more suitable and why another agent program is less suitable to support your answer. (1 + 4 marks)

[Total: 25 marks]

Question 2

- a) The *cutting stock problems*, as illustrated in **Figure 2-1**, occur when materials such as steel rods are processed in different lengths and diameters, among other things. It is a problem that aims to minimise material wasted, which arises in a variety of manufacturing industries such as rod industry, paper industry, wood industry, plastic, and leather manufacturing industry etc.

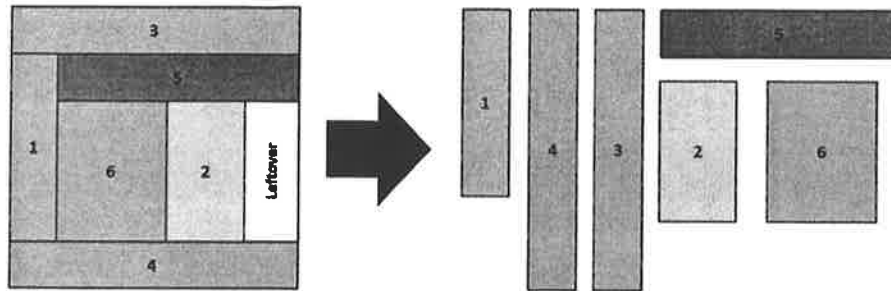


Figure 2-1: The cutting stock problems.

- (i) Determine whether the *cutting-stock problem* is an optimisation problem and justify your answer. (1 + 2 marks)
 - (ii) *Local search strategies* are usually used in solving optimisation problems. Identify **TWO (2)** differences between *local search strategies* and *classical search strategies*. (4 marks)
 - (iii) A *state-space landscape* is a graphical representation of different states of a problem reachable by a local search. Draw a *state-space landscape* that comprises **FIVE (5)** states usually encountered by a local search in which its elevation corresponds to a *heuristic cost function*. (4 marks)
- b) The *4-queens problem* is a problem of placing four chess queens on a 4×4 chessboard so that no two queens threaten each other; thus, a solution requires that no two queens share the same row, column, or diagonal. Suppose a one-column one-queen formulation is used to solve a problem as illustrated in **Figure 2-2**, where the heuristic cost function h is defined as *the number of pairs of queens that are attacking each other, either directly or indirectly*, answer the following three questions based on information provided in **Figure 2-2**:



Figure 2-2: An example of a 4×4 chessboard.

- (i) Calculate the number of successors for the problem. (1 mark)
- (ii) Calculate the h -value of the state of the problem provided in **Figure 2-2**. (1 mark)
- (iii) Redraw the chessboard provided in **Figure 2-2** by filling up all the unoccupied squares with the proper h -values. (12 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 3**

International Conference on Digital Transformation and Applications (ICDXA) is a conference organised by Tunku Abdul Rahman University of Management and Technology (TAR UMT) in 2024 that aims to promote research, education, and industry implementation that involve Digital Transformation (DX) processes. Suppose the conference will happen in two months' time, and you are assigned the task of allocating five events to three venues based on information provided in **Table 3-1** and **Table 3-2**, and each event will only be allocated one venue at a time, answer the following questions:

Table 3-1: Event.

No.	Event Code	Track	Time
1	P01-0001	Speech by industry speakers	9:00am – 12:00pm
2	P01-0002	Full paper presentation	10:00am – 1:00pm
3	P01-0003	Abstract presentation	11:00am – 2:00pm
4	P01-0004	Final year project exhibition	12:00pm – 3:00pm
5	P01-0005	Artificial Intelligence (A.I.) workshop	2:00pm – 5:00pm

Table 3-2: Availability of venues based on capacity.

No.	Lecturer	Events
1	Alan Turing	2, 3, 4
2	Steve Jobs	1, 2, 4, 5
3	Dennis Ritchie	1, 4, 5

- Formulate this problem as a *constraint satisfaction problem (CSP)* in which there is one *variable* per class, by stating *domains* for each variable (after the enforcement of unary constraints) and *constraints* in a format that represents *abstract relations*. (9 marks)
- Design a *constraint graph* associated with the *CSP*. (6 marks)
- Name the most popular algorithm applied for enforcing arc consistency. (1 mark)
- Identify domains of the variables after running arc-consistency on this initial graph based on the assumption that all unary constraints have already been enforced. (5 marks)
- Referring to the answer for **Question 3 d)**, identify the number of solutions available to this *CSP*. Then, write down **ONE (1)** solution. (1 + 1 marks)
- Identify whether your *CSP* solution for **Question 3 e)** is *consistent* or *inconsistent*, and whether it is *complete* or *partial*. (2 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4**

Entailment is an important concept frequently applied in the context of logical reasoning. For instance, in propositional logic, *entailment* is used to derive conclusions, via *model checking* and *theorem proving*.

- a) Briefly describe how *entailment* can be determined by *model checking*. (2 marks)

“Theorem proving is more efficient than model checking in determining entailment”.

Figure 4-1: A statement

- b) State your agreement with the statement provided in **Figure 4-1**. Provide **ONE (1)** reason to support your answer. (1 + 3 marks)
- c) Identify **THREE (3)** important concepts related to the implementation of theorem proving in determining entailment. Then, briefly describe each of them. (3 + 3 marks)

$$[\neg P \wedge (P \Rightarrow Q)] \Rightarrow \neg Q$$

$$P \vee [(\neg Q \wedge P) \wedge (Q \vee \neg P)]$$

Figure 4-2: Two sentences

- d) *Tautology* is a term which can be used interchangeably with **ONE (1)** of the three concepts provided as the answer for **Question 4 c)**. Determine whether the **TWO (2)** sentences provided in **Figure 4-2** are *tautology* or not with the use of *truth table*. (13 marks)

[Total: 25 marks]